

What Did Ten Points of Mean Recall Buy? Separating Group-Level from Case-Level Gains in Scene Graph Generation

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Two frozen models, one cell: relabelling within ϕ removes case identity and nothing else

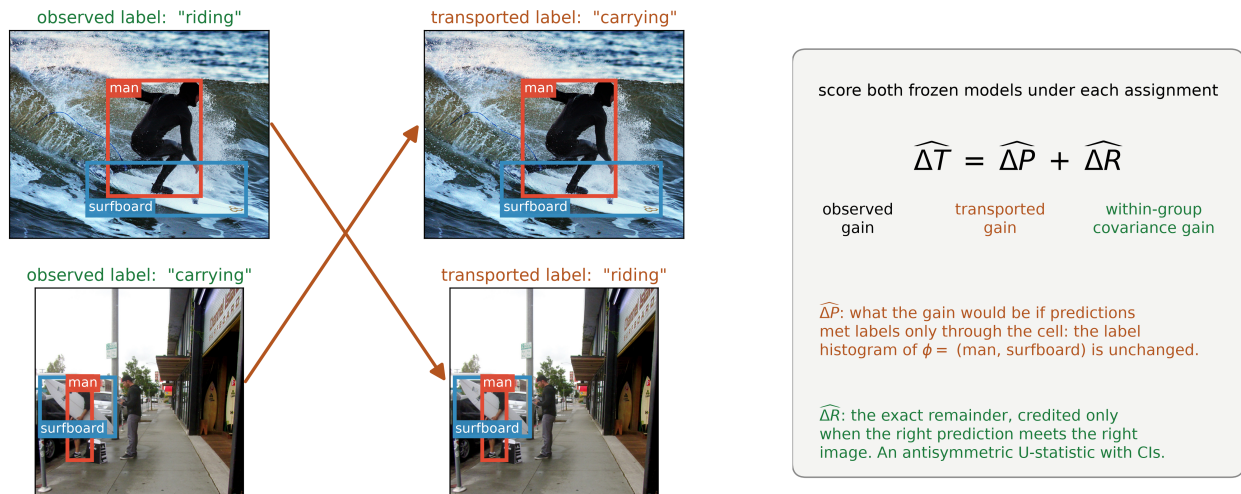


Figure 1. **Within-group relabelling.** Two Visual Genome relations share the subject–object pair (man, surfboard). Swapping their labels keeps the pair’s predicate histogram exactly as it was but breaks the link between each image and its own label. Scoring two frozen models under the observed and the swapped labels splits their score gain exactly: $\widehat{\Delta P}$ is the part that survives the swap — earned at the level of the group — and $\widehat{\Delta R}$ is the part that needs the right label on the right image.

Abstract

001 Debiasing methods for scene graph generation are judged
002 by mean recall, and one of the best known, Total Direct Ef-
003 fect (TDE), raises it on Visual Genome from 14.6 to 24.8.
004 What did those ten points buy? Mean recall cannot say. A
005 model raises it either by predicting rare predicates more of-
006 ten within each subject–object pair, or by recognising which
007 individual images actually show them, and the metric cred-
008 its both alike. We give an exact way to tell them apart. Scor-
009 ing each prediction against the label of a different relation
010 from the same subject–object group keeps the group’s pred-
011 icate frequencies but breaks the link between each relation
012 and its own label; this splits any proper-score difference
013 between two frozen models into a group-level part and a

case-level covariance, with no fitted baseline or tuning pa-
rameter, in $O(NK)$ time, with image-clustered confidence
intervals. On TDE’s released checkpoints with calibration
matched, the proper-score change is almost entirely group-
level: under the quadratic score the case-level part is indis-
tinguishable from zero, and under the log score it is small
and positive, under a tenth of the total. The converse also
occurs: a CLIP model reading image crops, ranked last
on accuracy and on the proper score, discriminates better
among relations sharing an object pair than the geometry
model ranked above it.

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1. Introduction

Scene graph generation is evaluated on a benchmark whose labels are unusually predictable from something every model sees. Given the ordered object pair (man, surfboard), the predicate on Visual Genome [15] is very often riding, and a lookup table of training frequencies is famously competitive [36]. Recall@ K rewards models that echo this prior, so the field now leads with mean recall, which averages recall over predicates and so rewards getting the rare ones [4, 31]. Debiasing methods are now judged by how far they raise it. Total Direct Effect (TDE) [32], among the most cited of them, raises mean recall@50 of a MOTIFS model from 14.6 to 24.8 on the canonical VG150 split, in our evaluation of the released checkpoint with the authors' own code.

What did those ten points buy? There are two ways to raise mean recall, and the metric cannot tell them apart. A model can move probability towards rare predicates *within each subject-object pair* — say carrying more often for (man, surfboard) — without becoming any better at deciding *which* images show a man carrying rather than riding. Or it can learn to make that decision: put more probability on the correct predicate for the individual image, beyond what the object pair implies. The first is a change in the model's group-level predictions; only the second is better recognition. Every existing remedy — head/body/tail-stratified recall [18], independent mean recall [20], zero-shot recall [22], rarity-stratified accuracy [14] — slices one model's performance differently. None attributes the gain *between* two models to one route or the other.

An exact split. We separate the two routes with one operation (Fig. 1). Take two frozen models and a test set grouped by a discrete feature both models observe, here the subject-object pair. Score each model's prediction for one relation against the label of a *different* relation in the same group. This relabelling preserves each group's predicate frequencies exactly, so any part of the score gain earned at the level of the group survives it; it destroys the correspondence between an image and its own label, so any part earned by getting individual cases right does not. The observed gain therefore splits exactly into a *group-level* part $\widehat{\Delta P}$ and a *case-level* remainder $\widehat{\Delta R}$, which is a within-group covariance between the change in the models' predictions and the label (Theorem 1). The split holds in the observed sample for any proper score, needs no fitted reference model, held-out fold or tuning parameter, costs $O(NK)$ on cached predictions, and comes with image-clustered intervals.

What it finds. On the two models of TDE's own released checkpoint, evaluated with the authors' code, and with each model's confidence matched on held-out im-

ages, the proper-score change is almost entirely group-level: the case-level part is under a tenth of it under either the quadratic or the log score, and under the quadratic score it is -0.00006 with an interval straddling zero (Sec. 5). The proper-score change behind the ten points is a change in which predicates the model favours within each pair, far more than in how well it tells that pair's relations apart. Nor is that implied by how TDE is built: in the released model its counterfactual subtraction cancels the image-specific visual logits as well as the frequency prior (Sec. 5), so it discards case-level evidence along with the group-level prior. The converse appears too. A model reading image crops through a frozen CLIP encoder [26], which every aggregate measure ranks last, gains $+0.00467$ of case-level covariance over the geometry-based model ranked above it, with calibration matched and an interval excluding zero (Sec. 6): its deficit is entirely group-level. That gain does not make it the better ranker, though, and the two measures' disagreement is itself part of what Sec. 6 reports.

Contributions.

- An audit of a published debiasing method's released checkpoints, finding its ten-point mean-recall gain backed by a proper-score change that is almost entirely group-level — a case-level part indistinguishable from zero under the quadratic score and under a tenth of the total under the log score — and a reversal in which the model ranked last is the best discriminator among relations sharing an object pair.
- An exact, model-free decomposition of the score gain between two frozen models into group-level and case-level parts, valid for every Bregman score, computed in one pass over cached predictions, with cluster-robust inference.
- Validation that the case-level part is exactly zero when it should be, covers at the nominal rate under image clustering, and separates calibration, prior and shortcut effects — together with a protocol for the one confound it does not separate.

What is not claimed. The split is conditional on the declared grouping: $\widehat{\Delta R}$ measures case-level discrimination *among relations sharing an object pair*, and credits any signal varying inside a pair, including a shortcut nobody recorded. It is not invariant to recalibration — rescaling a model's confidence moves score between the two parts while changing no ranking — which is why the comparisons whose models differ visibly in confidence, TDE's and the CLIP model's, are read confidence-matched. The split is of a proper score that weights every relation equally, whereas mean recall weights every predicate equally, so the audit speaks to the relation-weighted change; a case-level gain confined to rare predicates could be hidden by losses

on common ones. And the audit’s headline holds at the configuration we argue for, not at every configuration; Tab. 1 reports all five. These limits are taken up in Sec. 7.

2. Related work

Bias and its evaluation in scene graph generation.

Scene graph generation was posed as structured prediction over a whole image, solved by message passing between object and relation nodes [34]; MOTIFS then showed a frequency predictor of the predicate given the object pair to be surprisingly competitive [36]. Mean recall became the headline metric, and the causal TDE intervention was introduced, to reduce that dependence [32]; a long line of debiasing work has followed [16, 23], and the metrics themselves have been shown to reward pathologies [20], a diagnosis that recurs in open-vocabulary settings [19]. Benchmark audits outside scene graphs find the same pattern: VQA models collapse when answer priors change between train and test [1], GQA-OOD stratifies accuracy by answer rarity within question groups [14], and multimodal models exploit non-visual shortcuts [10, 37]. Long-tailed recognition faces the same question for re-weighting and logit adjustment [3, 24]. All of these characterise a *single* model. We attribute the gain *between* two.

Proper-score decompositions. Conditional on a grouping, a proper score splits into reliability and resolution terms [7, 25], for general proper scores [2]. Our case-level term is not that resolution term: it descends from the *covariance* decomposition of the Brier score due to Yates [35], which a monotone recalibration moves while the classical resolution is invariant to it. The two coincide only when both models are calibrated within every group. Variance estimators exist for single-model decompositions [30], but not for a contrast between two models scored on the same data.

Comparing two predictors. Inference on a paired score difference is the Diebold–Mariano test [9] and its conditional form [11]; at a single group our total-gain interval is a clustered Diebold–Mariano test (Corollary 3), so the decomposition adds a split to what those tests already report. DeLong *et al.* [8] compare correlated ROC areas with a paired U-statistic, the same device we use for a group-conditional score gain. The case-level term is an order-two U-statistic [13], with a Hájek projection supplying its influence function [29].

3. Splitting a score gain

Setup. Let $q_0(\cdot | x)$ and $q_1(\cdot | x)$ be the predictive distributions of an old and a new frozen model over K labels, and $S(q, y)$ a bounded proper score, larger being better. Our

default is the quadratic score $S(q, y) = 2q_y - \|q\|_2^2$, negative Brier loss up to a label-only constant; the log score is the second case we report. Let $C = \phi(X)$ be a discrete grouping both models observe, declared before the audit — for relation prediction, the ordered subject–object class pair. We call a value of C a *cell*. The gain vector

$$H_x(y) = S(q_1(\cdot | x), y) - S(q_0(\cdot | x), y), \quad y = 1, \dots, K, \quad (1)$$

can be evaluated at a counterfactual label without rerunning either model, and that is all the construction needs.

Transport. For a cell c , let $\Delta T_c = \mathbb{E}[H_X(Y) | C = c]$ be the observed gain and $\Delta P_c = \mathbb{E}_{H|c} \mathbb{E}_{Y'|c}[H_X(Y')]$ the gain when each case is scored against an *independent* label Y' drawn from the same cell, and set $\Delta R_c = \Delta T_c - \Delta P_c$. Transport keeps the cell and its label distribution and removes only the pairing between a case and its own label.

Theorem 1 (Transport and covariance identities). *For any integrable score contrast H ,*

$$\Delta T_c = \Delta P_c + \Delta R_c, \quad (2)$$

$$\Delta R_c = \sum_{y=1}^K \text{Cov}(H_X(y), \mathbb{1}\{Y = y\} | C = c). \quad (2)$$

For the quadratic score, with $\Delta q_y(X) = q_1(y | X) - q_0(y | X)$,

$$\Delta R_c = 2 \sum_{y=1}^K \text{Cov}(\Delta q_y(X), \mathbb{1}\{Y = y\} | C = c). \quad (3)$$

Consequently, if the gain vector is a function of c alone, $\Delta R_c = 0$.

Eq. (3) is what the split means. The new model earns case-level credit only insofar as its probability *change* rises on the label the individual case actually carries, relative to other cases in the same cell. Anything constant within a cell — shifting probability towards rare predicates for every (man, surfboard) relation alike — lands in ΔP , which is why we call it the group-level part. It is *not* simply “prior fit”: under the quadratic score ΔP_c is the improvement in how close the model’s mean forecast sits to the cell’s label frequencies *minus* the increase in the dispersion of its forecasts (Proposition 1). The covariance form is due to Yates [35] and generalises: for every Bregman score, including the log score, ΔR_c is the same covariance with Δq_y replaced by the contrast of the y -th gradient coordinate (Theorem 2). All proofs are in the supplementary.

Why it matters which part a gain is in. One cell, four cases, three labelled 1 and one labelled 2; the old model

predicts $(\frac{1}{2}, \frac{1}{2})$ for all four. Model A predicts $(\frac{3}{4}, \frac{1}{4})$ for every case — the cell’s frequencies. Model B predicts $(\frac{3}{4}, \frac{1}{4})$ for the three label-1 cases and $(\frac{1}{4}, \frac{3}{4})$ for the label-2 case: the same distance from the old model, but pointed at each case’s own label. Under the quadratic score A gains $+\frac{1}{8}$, all of it group-level, and B gains $+\frac{3}{8}$ with a case-level part of $+\frac{1}{2}$. Now let deployment see this cell’s labels in proportions $(\frac{1}{4}, \frac{3}{4})$ instead, a label shift. Reweighting each case by $p'(y)/p(y)$ sends A to $-\frac{3}{8}$ — worse than the model it replaced, with not one prediction changed — and leaves B at $+\frac{3}{8}$. The split said in advance which gain was exposed. With $m_y(z) = \mathbb{E}[H_X(z) \mid Y = y, C = c]$, the case-level part is a sum over label pairs, $\Delta R_c = \sum_{y < z} p(y)p(z) D(y, z)$ with $D(y, z) = m_y(y) - m_y(z) + m_z(z) - m_z(y)$, and a label shift leaves every D fixed and changes only the weights. With two labels, as here, a shift can rescale the case-level part but never reverse it; with three or more it can (Appendix C). A case-level gain is therefore less exposed to a shift than a group-level one, not immune to it. That is the operational reason to want them apart. (Appendix C works the example through.)

Estimator. In a cell with $n_c \geq 2$ cases, write $H_i(y) = H_{x_i}(y)$ and estimate

$$\widehat{\Delta T}_c = \frac{1}{n_c} \sum_i H_i(y_i), \quad \widehat{\Delta P}_c = \frac{1}{n_c(n_c-1)} \sum_{i \neq j} H_i(y_j), \quad (4)$$

$$\widehat{\Delta R}_c = \binom{n_c}{2}^{-1} \sum_{i < j} A_{ij}, \quad (5)$$

$$A_{ij} = \frac{1}{2} \{H_i(y_i) + H_j(y_j) - H_i(y_j) - H_j(y_i)\}. \quad (6)$$

These satisfy $\widehat{\Delta T}_c = \widehat{\Delta P}_c + \widehat{\Delta R}_c$ exactly in every sample, and $\widehat{\Delta R}_c$ is an unbiased order-two U-statistic [13] for ΔR_c (Theorem 3). Transporting a case only to the *other* cases’ labels matters: fitting the cell’s label frequencies on all n_c cases, the natural plug-in, shrinks every cell’s case-level gain by exactly $(n_c - 1)/n_c$, halving it at $n_c = 2$ (Proposition 4). With m_{cy} the count of label y in cell c , case i ’s transported score is

$$P_i = \frac{\sum_y m_{cy} H_i(y) - H_i(y_i)}{n_c - 1}, \quad (7)$$

so the whole estimator costs $O(NK)$ on cached predictions, with no fitted model, held-out fold or smoothing parameter. Cells are weighted by size; singletons cannot identify a within-cell counterfactual and are excluded and reported.

Inference. With $\bar{A}_i = (n_c - 1)^{-1} \sum_{j \neq i} A_{ij}$, the Hájek influence contribution of case i is

$$\widehat{\psi}_i = 2\bar{A}_i - \widehat{\Delta R}_c - \widehat{\Delta R}. \quad (8)$$

We sum $\widehat{\psi}_i$ within each image and use the one-way cluster-robust sandwich variance, since relations from one photograph are not independent. The estimator is asymptotically normal under bounded scores and non-dominating clusters (Theorem 4); at a single cell, the total-gain interval is a clustered Diebold–Mariano test [9] (Corollary 3). A randomization test that cyclically shifts labels within every cell is a secondary, relation-level diagnostic.

Confidence matching. One property of the split must be part of the procedure rather than a caveat. Because ΔR is a covariance between probability *movements* and labels, rescaling a fixed model’s confidence moves score between the two parts while changing no ranking, no top-1 decision and no information about any case (Sec. 4 measures how much). Two released checkpoints that differ in calibration therefore cannot be compared on their raw outputs. Wherever compared models differ visibly in confidence we split the images at random, fit each model a single temperature by maximum likelihood on one half, and run the split on the other half with each model’s own temperature applied. The protocol uses no label from the audited half.

Choosing the grouping. The split is conditional on ϕ . Merging cells changes it by an explicit between-cell covariance term of either sign (Proposition 2), so the finest grouping that the benchmark’s construction defends is the conservative choice. A relevant variable left out of ϕ is the same term with the fully adjusted cell as the fine partition, and Proposition 3 bounds it by a single elicited correlation, in the spirit of the robustness value of Cinelli and Hazlett [5].

4. Does the split measure what it claims?

A released checkpoint cannot validate the split, because what the model can see is unknown. We validate it where the answer is known, first in simulation and then on Visual Genome predictors whose information content we control. All intervals are 95%.

Simulation. We draw 24 cells over five labels with Dirichlet label distributions, take the old model to be exactly the cell prior, and let the new one interpolate towards an oracle. The estimated case-level gain grows linearly with the injected signal. Under a null that perturbs the prior predictions and reassigns them within cells, the randomization test rejects in 0.030 of 200 datasets at level 0.05, with mean estimate -8.97×10^{-5} . For the interval we use on real data, we simulate relations clustered into images with a shared per-image effect and Zipf-distributed cell sizes, most cells holding two cases: the image-clustered interval covers in 94.0% of 500 runs, against 88.6% for an interval that treats relations as independent.

Three designs check that the two parts separate what they claim to. A *prior-only* improvement, where the new model corrects a noisy estimate of the cell prior, gives a case-level gain centred at zero (0.0002, sd 0.0035). An *unrecorded within-cell shortcut* — a binary feature varying inside each cell that the new model uses — is credited to the case-level part (+0.041, sd 0.004): the split measures all within-cell signal relative to the declared grouping, shortcut or not. And a *calibration-only* change, where the new model is a temperature-softened copy of an overconfident one carrying identical case-level information, is credited a large spurious case-level *loss* of -0.488 on raw outputs, which the confidence-matching protocol of Sec. 3 reduces to -0.0003 (sd 0.0017). That last design is why the protocol is part of the procedure.

Controlled predictors on VG150. We train predicate classifiers whose inputs we control on a VG150-style split we reconstruct from the released Visual Genome annotations, with the same 150-object and 50-predicate vocabulary and images divided 70/30 between fitting and audit (Appendix N.1). It is not the canonical split of Sec. 5, which is why its test set is larger; models trained here from annotations need not reproduce a released split. The predictors are: a frequency table that sees only the class pair (FREQ), an MLP on class embeddings (MLP-CLASS), and MLPs adding twelve box-geometry features (MLP-SPATIAL, MLP-SPATIAL+). The audit covers 227,337 of 229,605 test relations, 99.0%, in 6,346 identified class-pair cells.

MLP-CLASS beats FREQ by 0.03386 of quadratic score and the split assigns *all* of it to the group-level part: the case-level gain is 0.00000 with a zero-width interval and randomization $p = .724$. That zero is algebraic, not empirical — both models are functions of the class pair alone, so Theorem 1 forces it — and it is the sharpest check that nothing is credited to case-level evidence when none exists. Adding box geometry then yields a case-level gain of 0.01439 (CI [0.01373, 0.01504], $p = .002$) that no aggregate score isolates. Against MLP-CLASS the total gain of 0.00885 even *understates* it, because the group-level part deteriorates by 0.00554: geometry moved the model’s probability towards each relation’s own label within its pair, and made it a slightly worse fit to the cell frequencies at once.

5. What TDE’s ten points bought

Tang *et al.* [32] release a causal MOTIFS PredCls checkpoint that yields two models from one set of weights: the baseline (MOTIFS-SUM) and its Total Direct Effect counterpart (MOTIFS-TDE). The baseline’s predicate logit is a sum of three terms: a visual term from the union region, a context term, and a frequency prior indexed by the object pair, which in PredCls is fixed by the subject-object cell.

Table 1. **TDE against its MOTIFS baseline** on VG150 PredCls, released checkpoints. Raw rows use every identified relation; matched rows fit each model’s temperature on a held-out half of the images and audit the other half. The interval on $\widehat{\Delta R}$ is image-clustered. Our reading is the matched quadratic row; all five are reported because the answer depends on the configuration.

	Score	ϕ	$\widehat{\Delta T}$	$\widehat{\Delta P}$	$\widehat{\Delta R}$ [95% CI]
raw	quad.	pair	−0.11651	−0.10296	−0.01355 [−0.01504, −0.01206]
raw	log	pair	+0.03277	+0.15897	−0.12619 [−0.13093, −0.12145]
raw	quad.	subject	−0.11536	+0.10175	−0.21711
matched	quad.	pair	−0.18264	−0.18258	−0.00006 [−0.00120, +0.00109]
matched	log	pair	−0.54760	−0.59157	+0.04396 [+0.04079, +0.04714]

TDE subtracts the same sum evaluated with an averaged context. In the released code the visual and frequency terms enter both halves of that subtraction unchanged, so they cancel, and TDE’s output is the context term minus its averaged counterpart. TDE therefore removes the frequency prior, which is constant within a cell, and also the image-specific visual term, which is not. We evaluate both with the authors’ own code on the canonical VG150 test split, so the two prediction sets are aligned relation for relation. The reproduction matches the published behaviour: the baseline reaches $R@50 = 0.6612$ with $mR@50 = 0.1459$, and TDE trades the former for the latter, reaching $mR@50 = 0.2476$ and raising zero-shot recall to $zR@50 = 0.1432$, while $R@50$ falls to 0.4588 and top-1 predicate accuracy from 0.6981 to 0.5577. Without the graph constraint the mean-recall gain reverses: no-graph-constraint $mR@50$ falls from 0.3260 to 0.2981. The ten points exist only under the graph-constrained protocol. The audit covers 183,639 relations over 26,446 images in 7,127 subject-object cells; 98.9% of the relations fall in the 5,044 cells holding at least two, and so are identified.

The released pair must be confidence-matched. The two models differ sharply in calibration: the temperatures fitted on held-out images are $T^* = 2.50$ for the baseline and 0.92 for TDE. Scored as they ship, they give a significant case-level *loss* for TDE (-0.01355 , Tab. 1), but Sec. 4 showed that a calibration difference alone manufactures exactly that kind of loss. We therefore read the matched rows.

Result. Once each model carries its own held-out temperature, TDE’s quadratic-score change is -0.18264 , of which -0.18258 is group-level. The case-level part is $\widehat{\Delta R} = -0.00006$, interval $[-0.00120, +0.00109]$, randomization $p = .555$: indistinguishable from zero (Fig. 2). Un-

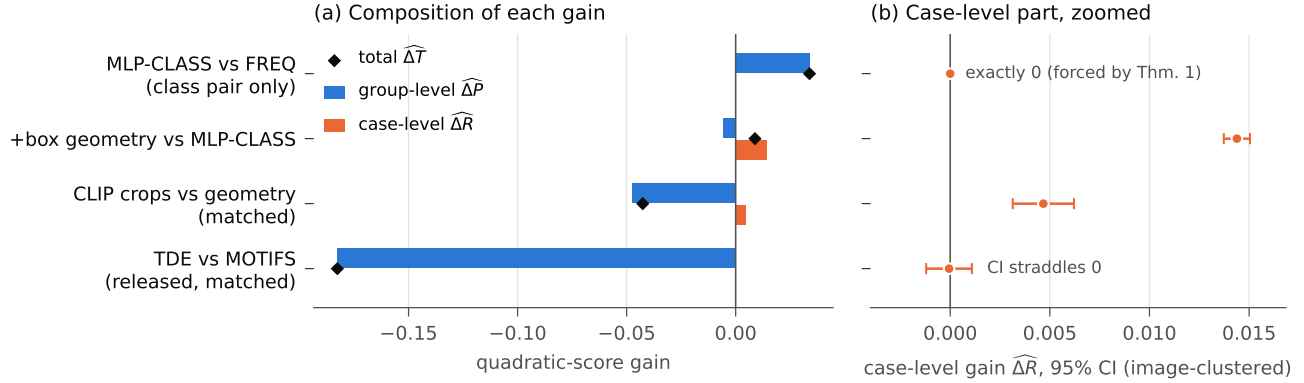


Figure 2. **Where each gain in this paper lives**, under the quadratic score. (a) Every comparison’s total gain $\widehat{\Delta T}$ (diamond) split into its group-level part $\widehat{\Delta P}$ and case-level part $\widehat{\Delta R}$; the two parts sum to the total exactly. (b) The case-level part alone with its image-clustered 95% interval. Two models built from the class pair alone differ by a purely group-level gain, as Theorem 1 forces. Box geometry and CLIP crops each buy a significant case-level gain, the CLIP model despite losing on the total. TDE’s large proper-score change is group-level, with a case-level part whose interval straddles zero. The CLIP and TDE rows, whose models differ visibly in confidence, are confidence-matched on held-out images.

der the log score the case-level part is small and positive, +0.04396 with an interval excluding zero, against a group-level change of -0.59157 . The two proper scores thus disagree on whether any case-level change survives matching, and agree that whichever is right it is small against the total: under a tenth of it even at the upper end of the log-score interval. Both are well-posed: each Bregman score defines its own exact estimand (Theorem 2), so choosing between them is a declaration and not an error to argue away. What both support is that TDE’s change to the predictive distribution is overwhelmingly group-level.

What that does and does not license. Mean recall rose ten points; the matched case-level gain is -0.00006 . Those are different units and we supply no bridge between them: it would be wrong to say ten points bought 0.00006 of evidence, and equally wrong to say the mean-recall gain is group-level, since the group-level part of the proper score fell. What the audit supports is a conjunction. The headline metric moved ten points, and the proper-score difference behind it lives almost entirely in how the model distributes probability across predicates within each object pair. Two cautions bound it. First, the proper score weights every relation equally and mean recall weights every predicate equally, so a case-level gain confined to rare predicates could be offset by a loss on common ones and vanish from the pooled split. Second, the result is not built into TDE. Its subtraction removes the frequency prior, a group-level term, but also the image-specific visual term, which carries case-level evidence; a case-level part near zero means whatever TDE lost there it recovered elsewhere, or neither moved, and the split as run here cannot say which.

The answer depends on the configuration. The five rows give case-level gains of -0.21711 , -0.12619 , -0.01355 , -0.00006 and $+0.04396$. At two of the five, “overwhelmingly group-level” is false: the raw log row has $|\widehat{\Delta R}| = 0.126$ against $|\widehat{\Delta P}| = 0.159$, and the subject-only row has -0.217 dominating $+0.102$. We argue for the matched class-pair row on stated grounds — a proper score punishes miscalibration, so calibration must be held fixed; and coarsening is not neutral (Proposition 2), so the finest grouping the benchmark defends is the conservative one — and report the rest so that a reader who rejects those grounds can see what follows instead.

6. The model every metric ranks last

The predictors of Sec. 4 never see an image: they read the class pair and, at most, box coordinates from the annotation. On the same reconstructed split we add one that reads pixels — subject and object regions cropped from the image and encoded by a frozen CLIP ViT-B/32 [26], concatenated with the class embeddings MLP-CLASS uses. CLIP is not fine-tuned, so any case-level gain reflects what pretraining had already recovered. All three compared models are retrained on one fixed subsample of 100,000 training relations, so differences isolate the features rather than the amount of data.

By every aggregate measure the CLIP model is the worst of the three: top-1 accuracy 0.6161 against 0.6467 for the geometry model, and a quadratic score 0.04655 lower. An evaluator reading either number would discard it. The split disagrees. With each model’s confidence matched on held-out images, the CLIP model loses 0.04727 of group-level score against the geometry model while *gaining* 0.00467 of case-level score (CI $[+0.00314, +0.00621]$); on raw outputs

the figures are 0.05296 and 0.00641 (CI [0.00514, 0.00769], $p = .002$). Its whole deficit is group-level (Fig. 2). Against the class-only model, pixels buy 0.01665 of case-level score on raw outputs where box geometry buys 0.01023, with well-separated intervals.

A larger case-level part is not the same as better ranking, and here the two disagree. With each model's prior matched within cells, the CLIP model still ranks the correct predicate first less often than the geometry model: top-1 accuracy 2.17 points lower, MRR 1.68 lower, recall@5 1.05 lower. Its probability moves towards the correct predicate within a pair more than the geometry model's does, but not by enough, often enough, to overtake it. So the CLIP model carries more case-level covariance without being the better ranker, and a calibration-free within-cell rank contrast is what would decide which describes recognition.

Two checks support this reading. Post-hoc correction of a model's label prior should move only the group-level part, and it does: +0.02401 group-level against +0.00111 case-level. Applied to the CLIP model it halves the aggregate deficit, from -0.04655 to -0.02143, while the case-level advantage persists (+0.00752; +0.00663 with both models corrected).

Beyond scene graphs. Nothing in Sec. 3 is specific to relations, and the supplementary applies the identical estimator to long-tailed image classification on CIFAR-100-LT across seeds and imbalance ratios, with superclasses as the grouping, and to long-tailed text classification. On CIFAR-100-LT two standard re-weighting schedules turn out to have opposite compositions once confidence is matched, reversing what their raw scores suggest about both. The constraint on the grouping bites here: the fine class label cannot serve as a cell, because it leaves every cell label-homogeneous and forces the case-level part to zero by construction — an artefact that reads exactly like a null result.

7. Limitations and conclusion

Limitations. Four bound what the split can say, and we would rather state them than have them found. *It is conditional on the grouping.* The case-level part measures discrimination among relations sharing an object pair; two equally fine partitions of the same cases can disagree completely about whether a gain is case-level (Appendix K), so declaring ϕ is a modelling decision that the benchmark, not the submitting author, should make before a leaderboard opens. *It credits any within-cell signal*, including shortcuts nobody recorded, and our own bound on that exposure is not reassuring: for the controlled-predictor pair of Sec. 4, an unrecorded covariate correlated at 0.061 with both parts would account for the entire case-level gain (Appendix I). *It is not invariant to recalibration*, which is why the comparisons whose models differ visibly in confidence

are read confidence-matched; a rank-based within-cell contrast would be invariant, at the cost of the exact additivity that is the point of the construction. And *it needs probabilities*: most benchmarks archive ranked predictions, and that, not compute, is the obstacle to routine use.

Conclusion. A ten-point mean-recall gain is two different achievements summed together, and the field's metrics report the sum. Scoring each prediction against another case's label from the same group separates them exactly, with no fitted reference model and with intervals that respect the clustering of relations within images. Applied to a released debiasing method, the split finds a large metric gain whose proper-score change is almost entirely group-level, with no detectable case-level change under the quadratic score; applied to a model every aggregate metric ranks last, it finds a case-level gain that the aggregates conceal. We suggest that benchmarks whose labels lean on a feature every model sees report the split beside the headline metric: the declared grouping and its coverage, both signed parts with the case-level interval, and the confidence-matched row. Code, the results files behind every number, and the scripts that regenerate the model predictions are in the supplementary material.

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